

Lab - Using Windows PowerShell (DAYUTH THY)

Objectives

The objective of the lab is to explore some of the functions of PowerShell.

Part 1: Access PowerShell console.

Part 2: Explore Command Prompt and PowerShell commands.

Part 3: Explore cmdlets.

Part 4: Explore the netstat command using PowerShell.

Part 5: Empty recycle bin using PowerShell.

Background / Scenario

PowerShell is a powerful automation tool. It is both a command console and a scripting language. In this lab, you will use the console to execute some of the commands that are available in both the command prompt and PowerShell. PowerShell also has functions that can create scripts to automate tasks and work together with the Windows Operating System.

Required Resources

- 1 Windows PC with PowerShell installed and internet access

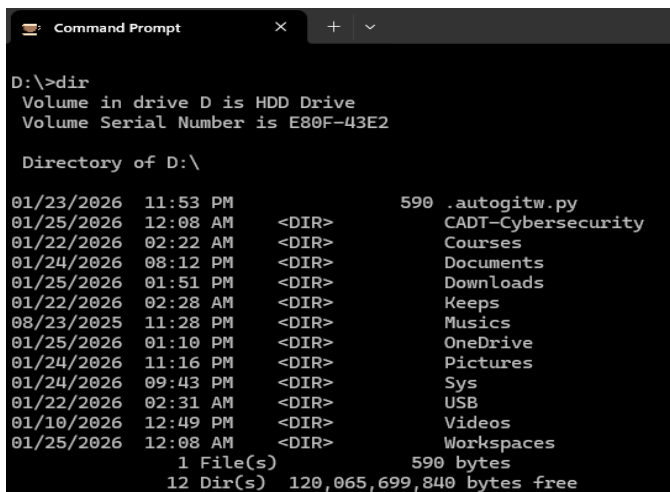
Instructions

Part 1: Access PowerShell console.

- Click **Start**. Search and select **powershell**.
- Click **Start**. Search and select **command prompt**.

Part 2: Explore Command Prompt and PowerShell commands.

- Enter **dir** at the prompt in both windows.
What are the outputs to the **dir** command?



```

Command Prompt
D:\>dir
Volume in drive D is HDD Drive
Volume Serial Number is E80F-43E2

Directory of D:\

01/23/2026  11:53 PM                590 .autogitw.py
01/25/2026  12:08 AM             <DIR> CADT-Cybersecurity
01/22/2026  02:22 AM             <DIR> Courses
01/24/2026  08:12 PM             <DIR> Documents
01/25/2026  01:51 PM             <DIR> Downloads
01/22/2026  02:28 AM             <DIR> Keeps
08/23/2025  11:28 PM             <DIR> Musics
01/25/2026  01:10 PM             <DIR> OneDrive
01/24/2026  11:16 PM             <DIR> Pictures
01/24/2026  09:43 PM             <DIR> Sys
01/22/2026  02:31 AM             <DIR> USB
01/10/2026  12:49 PM             <DIR> Videos
01/25/2026  12:08 AM             <DIR> Workspaces
               1 File(s)                590 bytes
               12 Dir(s) 120,065,699,840 bytes free
  
```

- b. Try another command that you have used in the command prompt, such as **ping**, **cd**, and **ipconfig**. What are the results?

```
D:\>ping google.com

Pinging google.com [172.217.25.46] with 32 bytes of data:
Reply from 172.217.25.46: bytes=32 time=38ms TTL=116
Reply from 172.217.25.46: bytes=32 time=27ms TTL=116
Reply from 172.217.25.46: bytes=32 time=29ms TTL=116
Reply from 172.217.25.46: bytes=32 time=128ms TTL=116

Ping statistics for 172.217.25.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 128ms, Average = 55ms
```

Part 3: Explore cmdlets.

- a. PowerShell commands, cmdlets, are constructed in the form of *verb-noun* string. To identify the PowerShell command to list the subdirectories and files in a directory, enter **Get-Alias dir** at the PowerShell prompt.

```
PS C:\Users\CyberOpsUser> Get-Alias dir
```

```
CommandType Name Version Source
```

```
Aliasdir -> Get-ChildItem
```

What is the PowerShell command for **dir**? Answer: Current Directory

```
PS C:\Users\t0fu> dir

Directory: C:\Users\t0fu

Mode                LastWriteTime         Length Name
----                -
d-----          1/1/2026   3:59 PM             .android
d-----         12/4/2025   12:54 AM          .arduinoIDE
d-----          1/22/2026    2:37 AM             .aws
d-----          1/22/2026    2:37 AM             .azure
d-----          1/22/2026    2:37 AM             .docker
d-----          1/21/2026   10:03 PM             .ssh
d-----          7/13/2025   12:52 AM          .th-client
d-----          1/25/2026    2:27 PM        .VirtualBox
d-----          7/12/2025    5:05 PM             .vscode
d-----          1/19/2026   11:30 PM Cisco Packet Tracer 8.2.2
d-r-----         1/21/2026    1:34 AM      Documents
d-r-----         1/17/2026    4:23 PM     Downloads
d-r-----          1/1/2026    4:18 AM     Favorites
d-r-----          1/1/2026    4:18 AM      Games
d-----         10/26/2025    3:11 PM      GNS3
d-----          1/25/2026    1:00 AM     GNS3 VM
d-----         10/5/2025    9:45 AM      go
d-r-----         11/16/2025   11:19 PM      Music
d-r-----          1/24/2026    2:16 PM     Pictures
d-r-----         11/17/2025    1:00 PM     Searches
d-r-----          1/23/2026    5:50 PM     Videos
d-----          1/24/2026    9:42 PM   VirtualBox VMs
-a-----          1/1/2026    2:00 PM    16 .emulator_console_auth_token
-a-----          8/20/2025    2:15 AM   154 .gitconfig
-a-----         12/20/2025   10:20 PM    20 .lessht
-a-----          1/19/2026   11:28 PM   174 .packettracer
```

- b. For more detailed information about cmdlets, perform an internet search for **Microsoft powershell cmdlets**.
- c. Close the Command Prompt window when done.

Part 4: Explore the netstat command using PowerShell.

- a. At the PowerShell prompt, enter **netstat -h** to see the options available for the **netstat** command.

```
PS C:\Users\CyberOpsUser> netstat -h
```

```
Displays protocol statistics and current TCP/IP network connections.
```

```
NETSTAT [-a] [-b] [-e] [-f] [-n] [-o] [-p proto] [-r] [-s] [-x] [-t] [interval]
```

```
-a Displays all connections and listening ports.
```

```
-b Displays the executable involved in creating each connection or listening port.
In some cases well-known executables host multiple independent components, and in
these cases the sequence of components involved in creating the connection or
listening port is displayed. In this case the executable name is in [] at the bottom,
on top is the component it called, and so forth until TCP/IP was reached. Note that
this option can be time-consuming and will fail unless you have sufficient
permissions.
```

```
<some output omitted>
```

- b. To display the routing table with the active routes, enter **netstat -r** at the prompt.

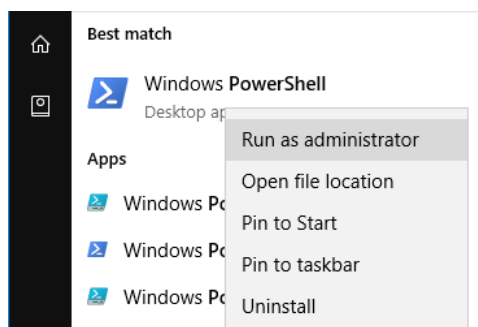
```
PS C:\Users\t0fu> netstat -r
=====
Interface List
21...00 ff f9 aa 6c aa .....TAP-Windows Adapter V9 for OpenVPN Connect
20...0a 00 27 00 00 14 .....VirtualBox Host-Only Ethernet Adapter
8.....OpenVPN Data Channel Offload
12...30 9c 23 aa 74 c4 .....Intel(R) Ethernet Connection (7) I219-V
19...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
7...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
10...00 50 56 c0 00 13 .....VMware Virtual Ethernet Adapter for VMnet19
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.0.1      192.168.0.167    35
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.0.0                255.255.255.0    On-link          192.168.0.167    291
192.168.0.167              255.255.255.255  On-link          192.168.0.167    291
192.168.0.255              255.255.255.255  On-link          192.168.0.167    291
192.168.1.0                255.255.255.0    On-link          192.168.1.10     281
192.168.1.10               255.255.255.255  On-link          192.168.1.10     281
192.168.1.255              255.255.255.255  On-link          192.168.1.10     281
192.168.120.0              255.255.255.0    On-link          192.168.120.1    291
192.168.120.1              255.255.255.255  On-link          192.168.120.1    291
```

What is the IPv4 gateway?

```
IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.0.1      192.168.0.167    35
```

- c. Open and run a second PowerShell with elevated privileges. Click **Start**. Search for PowerShell and right-click **Windows PowerShell** and select **Run as administrator**. Click **Yes** to allow this app to make changes to your device.



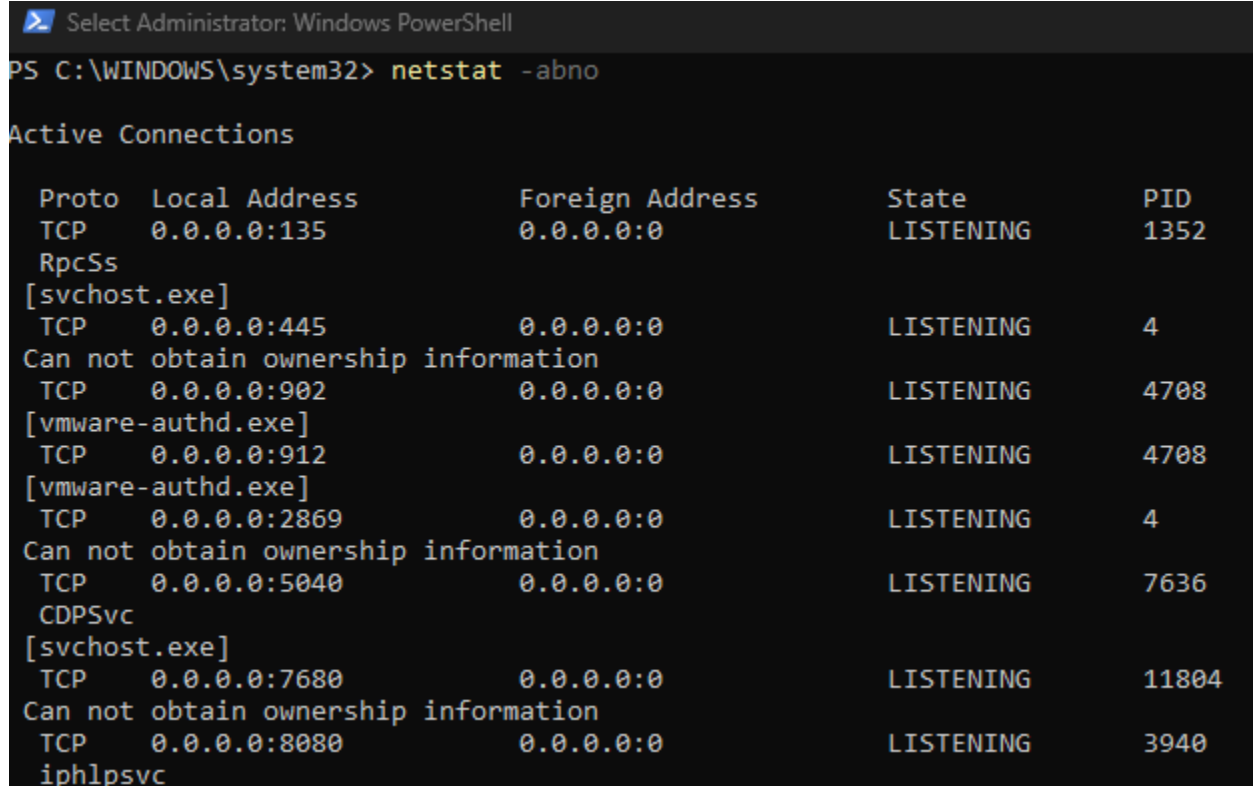
- d. The netstat command can also display the processes associated with the active TCP connections. Enter the **netstat -abno** at the prompt.

```
PS C:\Windows\system32> netstat -abno
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	756
RpcSs				
[svchost.exe]				
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
Can not obtain ownership information				
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	444
Can not obtain ownership information				
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	440
Schedule				
[svchost.exe]				
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	304
EventLog				
[svchost.exe]				
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	1856
[spoolsv.exe]				
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	544

<some output omitted>



The screenshot shows a Windows PowerShell window titled "Select Administrator: Windows PowerShell". The command prompt shows the command `netstat -abno` being executed. The output displays active connections, including listening ports and the processes associated with them. The output is as follows:

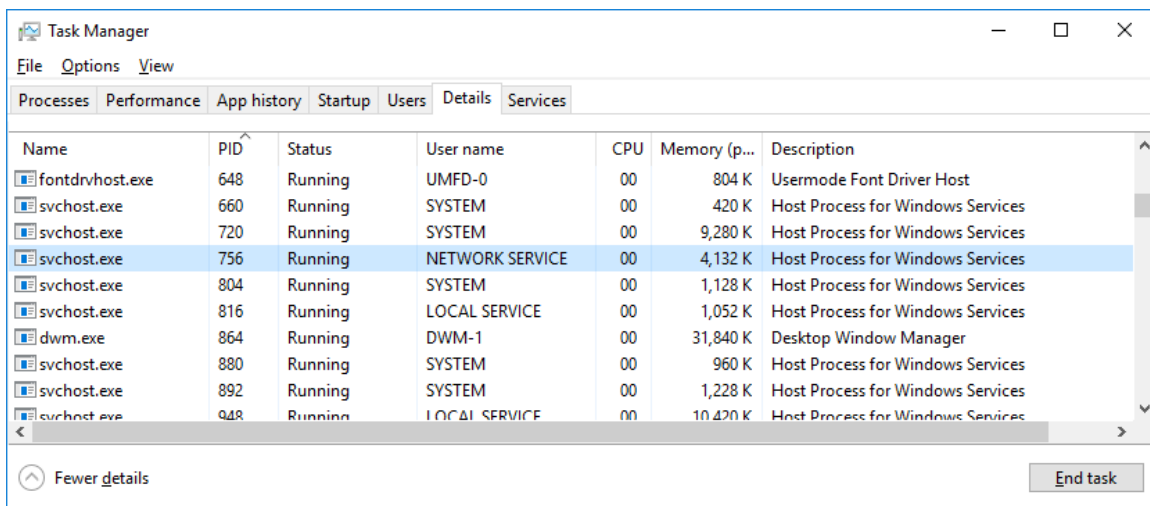
```
PS C:\WINDOWS\system32> netstat -abno

Active Connections

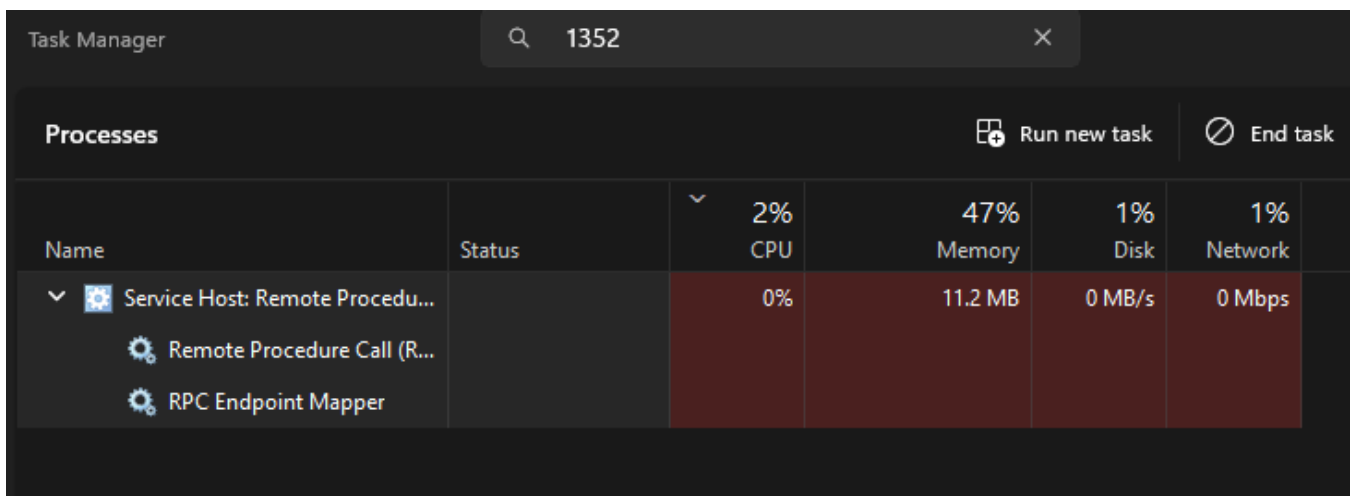
 Proto Local Address           Foreign Address         State           PID
----
TCP    0.0.0.0:135             0.0.0.0:0              LISTENING      1352
RpcSs
[svchost.exe]
TCP    0.0.0.0:445             0.0.0.0:0              LISTENING      4
Can not obtain ownership information
TCP    0.0.0.0:902             0.0.0.0:0              LISTENING      4708
[vmware-authd.exe]
TCP    0.0.0.0:912             0.0.0.0:0              LISTENING      4708
[vmware-authd.exe]
TCP    0.0.0.0:2869            0.0.0.0:0              LISTENING      4
Can not obtain ownership information
TCP    0.0.0.0:5040            0.0.0.0:0              LISTENING      7636
CDPSvc
[svchost.exe]
TCP    0.0.0.0:7680            0.0.0.0:0              LISTENING      11804
Can not obtain ownership information
TCP    0.0.0.0:8080            0.0.0.0:0              LISTENING      3940
iphlpvc
```

- e. Open the Task Manager. Navigate to the **Details** tab. Click the **PID** heading so the PID are in order.

- f. Select one of the PIDs from the results of `netstat -abno`. PID 756 is used in this example.
- g. Locate the selected PID in the Task Manager. Right-click the selected PID in the Task Manager to open the **Properties** dialog box for more information.



What information can you get from the Details tab and the Properties dialog box for your selected PID?



Part 5: Empty recycle bin using PowerShell.

PowerShell commands can simplify management of a large computer network. For example, if you wanted to implement a new security solution on all servers in the network you could use a PowerShell command or script to implement and verify that the services are running. You can also run PowerShell commands to simplify actions that would take multiple steps to execute using Windows graphical desktop tools.

- a. Open the Recycle Bin. Verify that there are items that can be deleted permanently from your PC. If not, restore those files.
- b. If there are no files in the Recycle Bin, create a few files, such as text file using Notepad, and place them into the Recycle Bin.
- c. In a PowerShell console, enter **clear-recyclebin** at the prompt.

```
PS C:\Users\CyberOpsUser> clear-recyclebin
```

Confirm

Are you sure you want to perform this action?

Performing the operation "Clear-RecycleBin" on target "All of the contents of the Recycle Bin".

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): y

What happened to the files in the Recycle Bin? Answer: Gone

```
PS C:\Users\t0fu> clear-recyclebin
```

Confirm

Are you sure you want to perform this action?

Performing the operation "Clear-RecycleBin" on target "All of the contents of the Recycle Bin".

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"):

```
PS C:\Users\t0fu>
```

Reflection Question

PowerShell was developed for task automation and configuration management. Using the internet, research commands that you could use to simplify your tasks as a security analyst. Record your findings.

It use very detail and deep dive in the Powershell command lines which can grab any detail of the OS to the terminal both very powerful too.